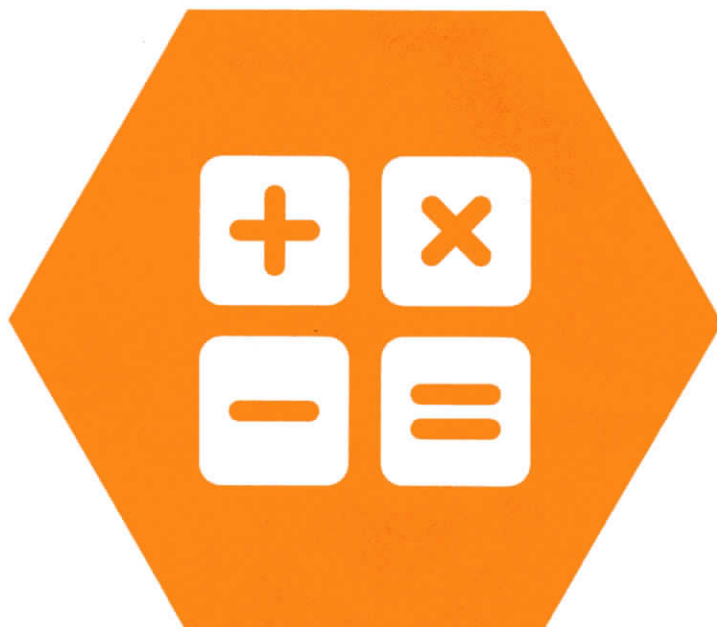




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
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PAPER A



2014 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

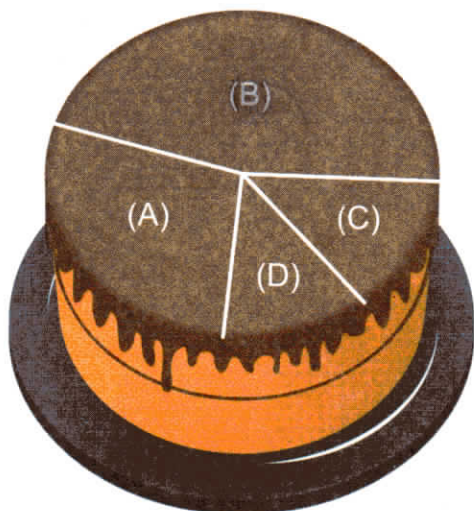
You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.
You are **NOT** allowed to use a calculator.

1. Amy cut a cake into four unequal pieces. Which piece was the largest?



2. This is the start of a number pattern.

12, 14, 16, 18, 20

What are the next two numbers in the pattern?

- (A) 21, 22
- (B) 22, 24
- (C) 22, 26
- (D) 30, 40

3. Which of these is **NOT** a triangle?



(A)



(B)



(C)



(D)

4. Which of these is the number eighty three?

- (A) 803
- (B) 83
- (C) 80
- (D) 38

5. $54 + 23 =$?

- (A) 59
- (B) 75
- (C) 77
- (D) 86

6. Sam drew a picture of a bee.



Which of these is also a picture of Sam's bee?



(A)



(B)



(C)



(D)

7. This chart shows the shells some children collected at the beach.

Name	Shells
Peter	
Heidi	
Sue	
Henry	

How many shells did Sue collect altogether?

- (A) 4
- (B) 5
- (C) 6
- (D) 8

8. $24 - 9$ is _____ $8 + 9$

Which of these makes the statement true?

- (A) bigger than
- (B) less than
- (C) equal to
- (D) half of

9. These clocks show a pattern of times.



Which of these clocks must replace the  to fit the pattern?



10. Mr Merlino measured the area of his garden.



Which unit should Mr Merlino use?

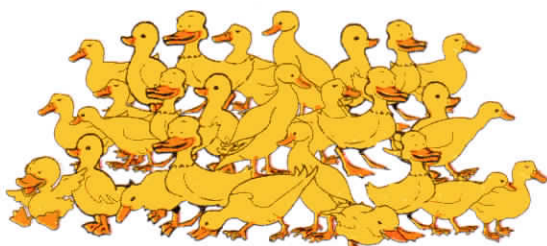
- (A) millilitres
- (B) kilograms
- (C) centimetres
- (D) square metres

11. Toby has 21 apples. Lillian has 4 apples less than Toby.

How many apples does Lillian have?

- (A) 4
- (B) 17
- (C) 18
- (D) 25

12. Tala had 28 ducklings in a pen. She took them out for a swim in groups of five until there were less than five left in the pen.



How many ducklings were left in the pen?

- (A) 3
- (B) 4
- (C) 5
- (D) 23

13. Lara has a robot called Speedy. Speedy sits in the middle square of a grid.

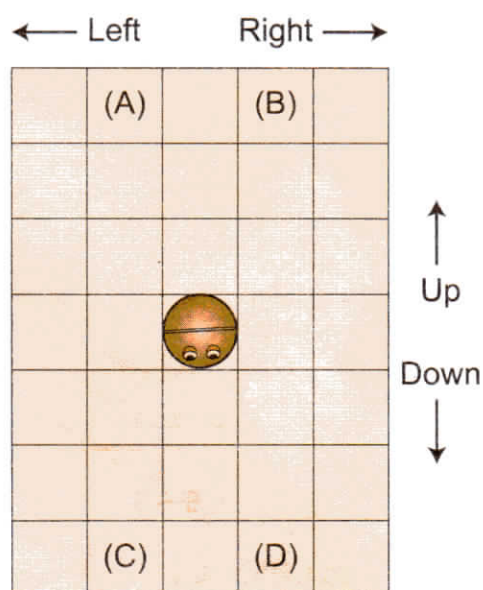
Speedy follows instructions for movement:

- U1 means move up 1 square
- D2 means move down 2 squares
- R1 means move to the right 1 square
- L2 means move to the left 2 squares

Lara gave these instructions to Speedy:

R1, D3, L2

Where will Speedy be after he has followed these instructions?



14. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15
- (B) 16
- (C) 31
- (D) 56

15. This shape has been made by joining two regular hexagons and one square.

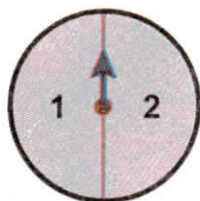


The perimeter of each hexagon is 12 centimetres.

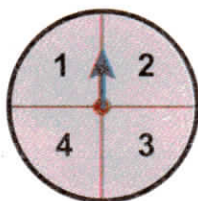
What is the perimeter of the whole shape?

- (A) 12 centimetres
- (B) 24 centimetres
- (C) 28 centimetres
- (D) 32 centimetres

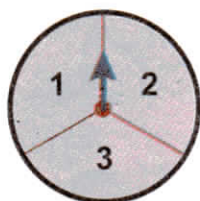
16. Which spinner has the greatest chance of landing on 2?



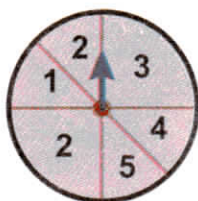
(A)



(B)



(C)



(D)

17. Here are two views of the same solid.



top view

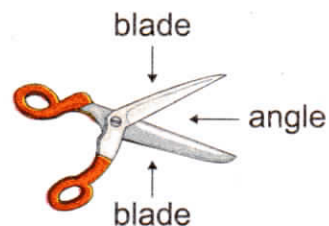


side view

What is the name of this solid?

- (A) cylinder
- (B) sphere
- (C) prism
- (D) cone

18. The blades of scissors can make an angle.



Which of these has the largest angle between its blades?



(A)



(B)



(C)

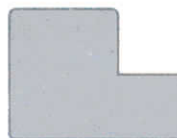


(D)

19. Anish drew this shape.



Which option shows the shape after a quarter turn clockwise?



(A)



(B)



(C)

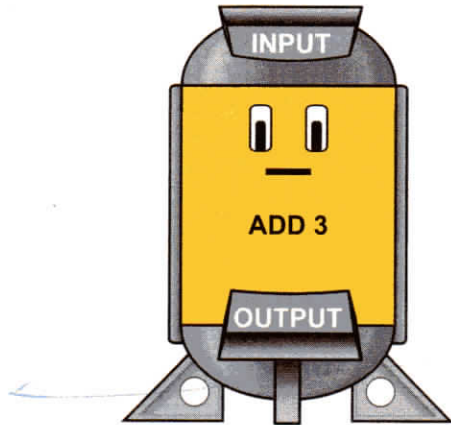


(D)

20. Ms Green made a number machine.

This machine adds 3 to any number put into it.

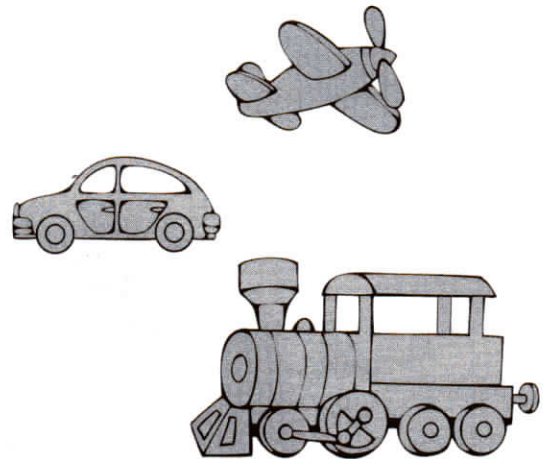
Carl put only odd numbers through the machine.



What kind of numbers would come out of the machine?

- (A) odd numbers only
- (B) even numbers only
- (C) the number 6 only
- (D) some even and some odd numbers

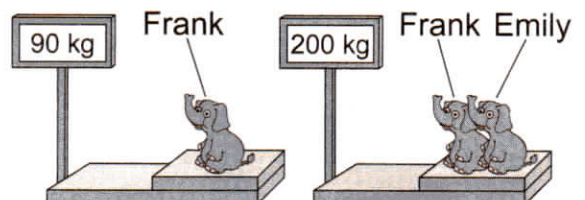
21. Max is painting some wooden toys. He can choose red, yellow or green. Each toy must be painted with a different paint.



How many different ways can Max paint his toys?

- (A) 12
- (B) 9
- (C) 6
- (D) 3

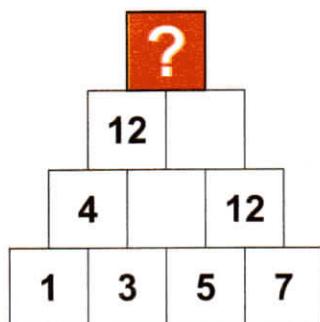
22. Emily and Frank are two baby elephants.



What is Emily's mass?

- (A) 90 kilograms
- (B) 110 kilograms
- (C) 190 kilograms
- (D) 200 kilograms

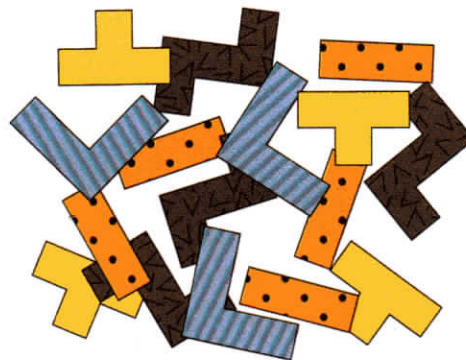
23. Sandy made a number pattern. She started on the bottom row. She added two numbers that were next to each other and wrote the answer in the box above. Some of the numbers are missing.



What must the value of ? be?

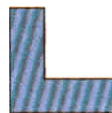
- (A) 22
- (B) 24
- (C) 32
- (D) 44

24. Andy scatters some puzzle pieces on the floor.



He picks one puzzle piece without looking.

Which piece is Andy **least** likely to pick?



(A)



(B)



(C)



(D)

25. Emma lives in Wonderland. In Wonderland one dollar equals 8 cents.

Emma bought a toy that cost 16 Wonderland dollars.

How much did the toy cost in Wonderland cents?

- (A) 2
- (B) 14
- (C) 24
- (D) 128

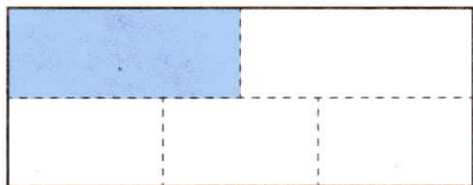
26. This is a number pattern.

209
228
247
266
285
?
323
342
361
380

What is the missing number?

- (A) 294
(B) 296
(C) 304
(D) 306

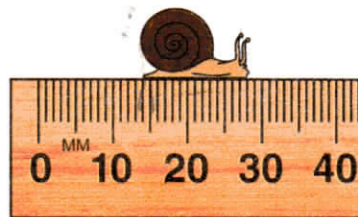
27.



What fraction of this shape is shaded?

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{1}{5}$ (D) $\frac{1}{3}$

28. A snail was crawling along a ruler.



Fully stretched, the end of the snail's tail was on the 14 mm mark and its head just reached the 28 mm mark on the ruler.

Then the snail crawled to the right and stretched itself again. The end of its tail was now on the 18 mm mark.

What mark on the ruler did the snail's head reach?

- (A) 24 mm
(B) 32 mm
(C) 34 mm
(D) 36 mm

29. Jess started listening to a CD when the minute hand on the clock was pointing to the 9.

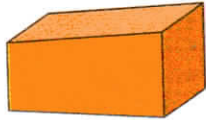
When she finished listening to the CD, the minute hand on the clock was pointing to the 5.



How long could Jess have listened to the CD?

- (A) 40 minutes
(B) 20 minutes
(C) 14 minutes
(D) 4 minutes

30. Which of these objects is **NOT** a prism?



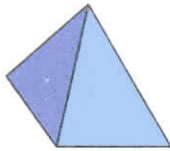
(A)



(B)

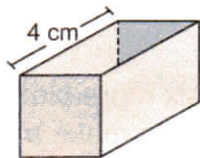


(C)

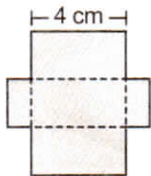


(D)

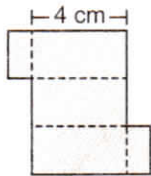
31. This picture shows a box without a lid.



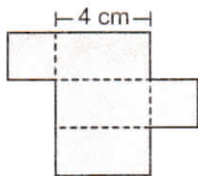
Which of these shows the net for this box?



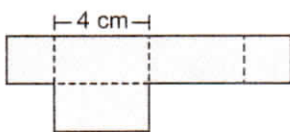
(A)



(B)

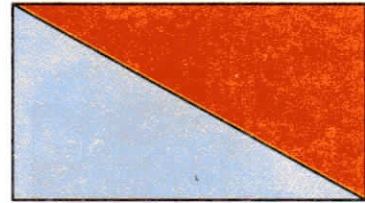


(C)



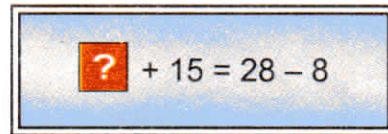
(D)

32. How many lines of symmetry does this picture have?



- (A) 4
- (B) 2
- (C) 1
- (D) 0

33.



What is the missing number?

- (A) 5
- (B) 10
- (C) 20
- (D) 35

34. Each week Natalia receives \$30 pocket money and saves one-third of it.

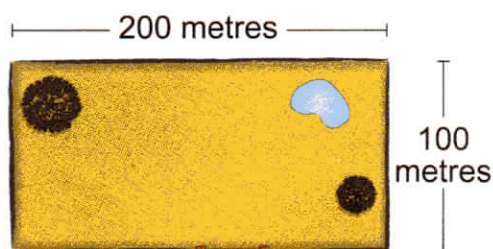
Each week Tony receives \$16 pocket money and saves half of it.

In eight weeks, how much money do Natalia and Tony save altogether?

- (A) \$368
- (B) \$144
- (C) \$88
- (D) \$18

35. Gary the farmer has two fields.

One field is in the shape of a rectangle.



The other field is in the shape of a square.

The perimeter of the rectangular field is 400 metres larger than the perimeter of the square field.

What is the length of the side of the square field?

- (A) 50 metres
- (B) 100 metres
- (C) 150 metres
- (D) 250 metres

36. Tara is adding two numbers. Some of the digits are hidden by symbols.

$$\begin{array}{r} \star \square \\ + 4 \star \\ \hline 103 \end{array}$$

The two digits hidden by stars are the same.

What digit is hidden by the red square?

- (A) 1
- (B) 5
- (C) 7
- (D) 8

37. Abi is 1 year and 9 months older than Ben.

Ben is 2 years and 8 months younger than Cam.

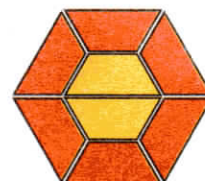
What is the difference in age between Abi and Cam?

- (A) 11 months
- (B) 13 months
- (C) 4 years and 5 months
- (D) 4 years and 7 months

38. Tom had some blocks that were all the same size and shape. He used two of them to make this regular hexagon.



He placed six more blocks around this hexagon to make a bigger regular hexagon.



How many more blocks does he need to place around this shape to make the next bigger regular hexagon?

- (A) 6
- (B) 10
- (C) 12
- (D) 18

39. Mr Lee makes jewellery. He makes two different kinds of bracelets. The Supa bracelet has three star beads and four oval beads. The Glowy bracelet has five star beads and three oval beads.



Supa



Glowy

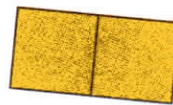
Mr Lee made 25 bracelets. Altogether he used 101 star beads.

How many Glowy bracelets did he make?

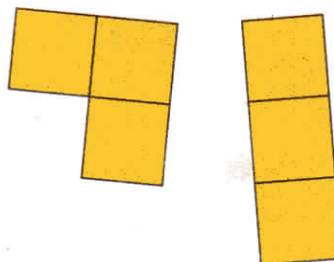
- (A) 20
- (B) 16
- (C) 13
- (D) 12

40. Finn has some square tiles.

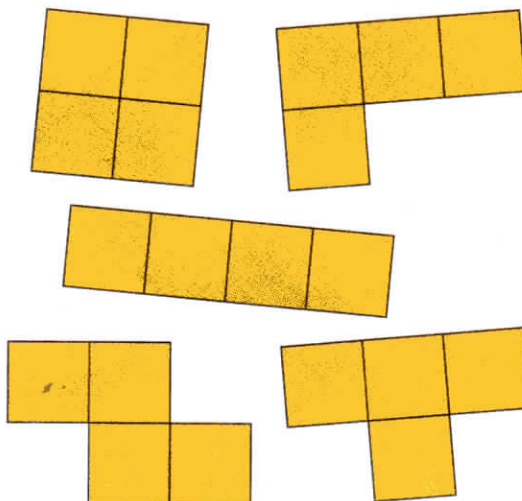
He uses two tiles together. He can only make this shape.



He uses three tiles together. He can only make these two different shapes.



He uses four tiles together. He can only make these five different shapes.



How many different shapes can Finn make using 5 tiles?

- (A) 6
- (B) 10
- (C) 12
- (D) 16

Acknowledgment

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The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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M | 2014 Mathematics Answer Keys

ICAS

International Competitions
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
1	B	D	B	D	A	A	B	A
2	B	B	C	A	A	A	C	B
3	C	D	B	A	D	C	B	C
4	B	B	D	C	B	D	A	B
5	C	C	C	A	C	D	C	C
6	A	B	A	C	D	B	C	A
7	C	C	D	A	A	C	A	A
8	B	A	C	B	D	A	D	B
9	C	D	A	D	A	B	C	C
10	D	C	D	B	D	B	A	B
11	B	B	C	D	D	A	B	C
12	A	B	B	A	B	B	D	D
13	C	C	D	D	D	B	A	D
14	C	D	D	C	B	C	D	A
15	B	C	B	C	B	C	B	B
16	A	D	D	D	C	B	A	B
17	D	A	A	B	B	C	D	C
18	D	C	B	A	A	B	B	D
19	B	A	D	A	B	C	B	D
20	B	C	A	D	C	D	A	D
21	C	B	C	B	B	C	A	B
22	B	D	D	A	D	A	C	A
23	C	C	C	C	C	A	C	C
24	A	B	A	C	B	B	B	A
25	D	D	B	B	C	D	C	B
26	C	C	B	D	D	C	D	A
27	B	D	C	C	C	B	B	D
28	B	A	A	C	C	A	D	A
29	A	A	D	D	A	D	B	D
30	D	B	C	B	A	C	B	B

(Please turn over)



M

2014 Mathematics Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Papers G & H	Papers I & J
31	C	D	D	A	C	A	C	C
32	D	D	C	B	C	B	A	B
33	A	A	A	B	A	C	D	D
34	B	B	C	D	D	B	A	D
35	A	D	B	A	B	C	B	C
36	D	D	C	82 082	36 036	327	120	120
37	A	C	B	21 021	13 013	256	41 041	864
38	B	A	C	92 092	327	21 021	83 083	478
39	C	D	B	125	12 012	640	864	540
40	C	C	C	24 024	33 033	41 041	45 045	36 036

Contact Details

Telephone: +61 2 8344 1016
Email: info@eaa.unsw.edu.au
Website: www.eaa.unsw.edu.au